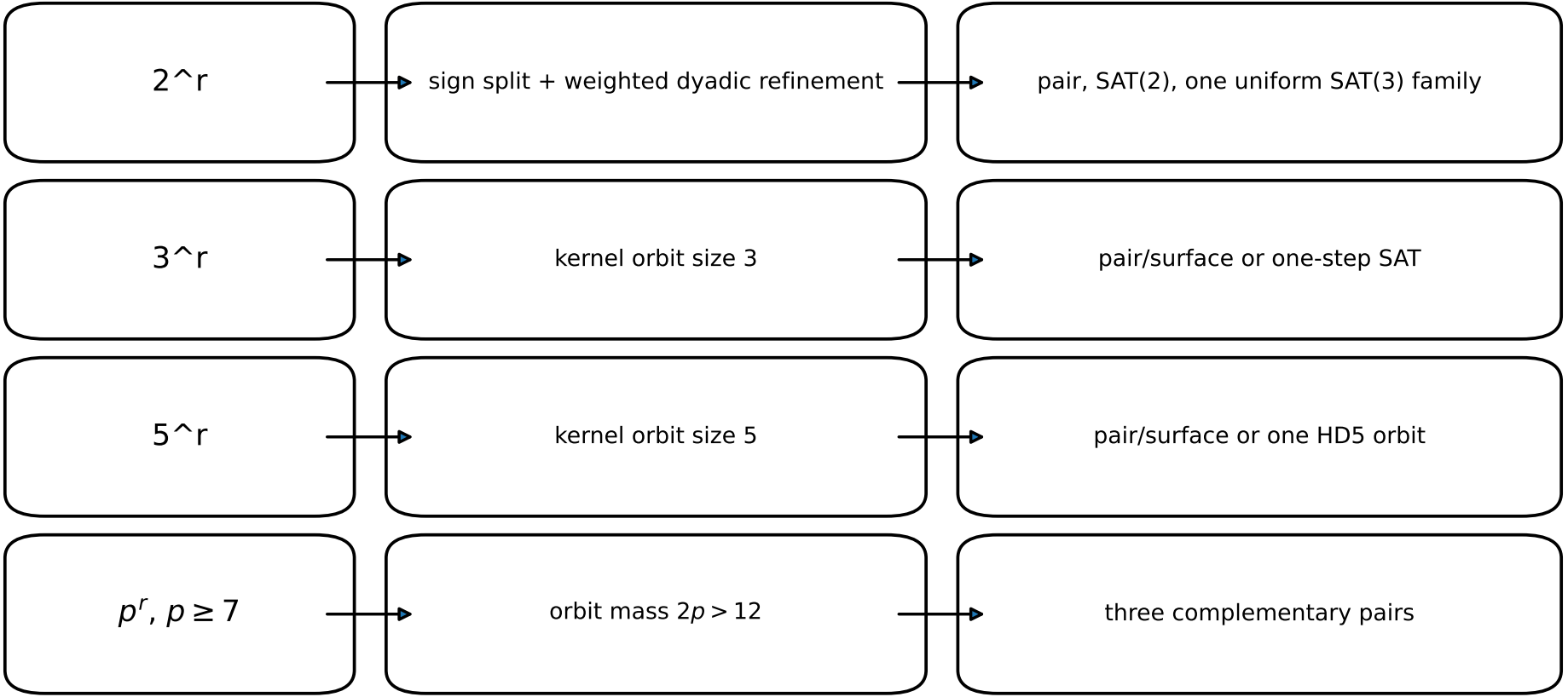Prime-power packet classification